varquanMS: local-scale, adaptive-precision archives for single-integration Measurement Sets

Working methods and experiment paper — 23 October 2026. This document describes implemented behavior and measured results, not an approved archival science profile. All sizes use decimal MB unless explicitly stated.

Abstract

We investigate an independently decodable visibility archive that preserves Measurement Set (MS v2) coordinates and auxiliary information while quantizing complex visibilities in local baseline–frequency blocks. Each block uses one stored float32 scale, exact exception values, and a measured choice among 4, 8, 12 and 16 bits per real/imaginary component or exact float32 components. Precision is selected by encoded size subject to reconstruction-error constraints, not by an amplitude-to-bit lookup. Baseline ordering changes which spectra share a scale; stored permutations recover original row, channel and correlation order.

The inspected test MS has 62,128 rows, 192 channels, four linear-feed products, and one integration. Its physical file total is 409.595 MB, including 381.714 MB of logical complex64 DATA. A measured amplitude-sorted, 128-baseline × 192-channel configuration produces a 106.730 MB archive, a **3.838× whole-MS reduction**. Mean per-component visibility MSE is `2.8117e-8` in stored DATA units squared. Matched WSClean 2048^2, 4-arcmin, multiscale imaging gives dirty/restored relative RMS differences of **0.1996% / 0.7137%** over the valid celestial footprint. These image metrics are not thermal-noise estimates. Three sequential encoding runs with 24 CPU workers take **12.19 s median**, range **11.71–12.66 s**, for a 106.731 MB archive. The benchmark includes startup, inspection, writing and integrity checks; verification, MS export and imaging are timed separately. It uses a shared host with warm/unspecified cache. Section 9 describes the test conditions, individual runs and output validation.

A separate 32×24 two-level MX experiment uses an outer float32 factor and 32-real-component micro scales. Under the tested bounds, 89.65% of components require a custom scaled-FP16 format. A tighter-error uniform control is 19.50% smaller in serialized component bytes with 7.34× lower component MSE; this MX configuration does not improve the measured visibility rate–distortion tradeoff.

A corrected full-range unsigned codebook uses all 2^b codes at the cost of losing the zero level. At matched grouping and error ceilings its full-MS component MSE is 0.223% lower, but whole-archive size is essentially unchanged at 106.732 MB. Median encoding time is 13.04 s in a new three-repeat comparison. Zero-rich autocorrelations acquire imaginary bias, so the codebook remains optional and the zero-preserving mapping remains the default.

1. Scope and relationship to earlier work

The goal is to reduce archive size without time/channel averaging or regeneration of original coordinates. MS v2 associates visibility cells with row identities, subtable references, flags and weights. UVW is explicitly a three-component double-precision coordinate in metres. Reconstruction therefore restores stored UVW rather than recomputing it. MS v2 specification

Normalization and quantization are not new contributions. Dysco combines grouped normalization, nonlinear quantization and dithering, including antenna/frequency factorizations, and evaluates image-domain errors. Its published telescope results are dataset-specific, not pre-

dictions for this MS. This prototype investigates adaptive precision, grouping controls, explicit exceptions/error accounting and standalone reconstruction; no matched Dysco benchmark has yet been run. Offringa (2016)

QLoRA motivates engineering comparisons such as distribution-adapted codebooks and quantized scales, but neither language-model quality nor NF4's design is a visibility-fidelity guarantee. The present codec is uniform, without NF4 or scale quantization in the production archive path. Section 10 separately evaluates experimental MX floating-point elements. Dettmers et al. (2023) XRADIO/MS v4 illustrates labeled-array schemas and homogeneous configurations; this archive does not claim MS v4 conformance and exports MS v2. XRADIO overview

2. Inspected observation and storage accounting

The source is `20260904_190008_59MHz.ms`, copied from the supplied 59 MHz event directory and checked against the source tar. Measurements below refer to the extracted MS, not its tar wrapper. The channel axis is **inside each DATA cell**; it is not 192 separate single-channel records. The full machine-readable inventory is initial-inspection.json.

Property	Inspected value
Integration / data description / spectral window / field counts	1 / 1 / 1 / 1
Antennas; rows	352; 62,128
Autocorrelation / cross-correlation rows	352 / 61,776
Python DATA shape and dtype	`[62128, 192, 4]`, complex64
Correlation order read from metadata	XX, XY, YX, YY
Complex sample count	47,714,304
Channel-centre range	59.3359375–63.90576171875 MHz
Channel width	23,925.78125 Hz
TIME; TIME_CENTROID	5295265208.539846; 5295265213.555357 s, UTC
UVW dtype / units / recorded reference	float64 / m / ITRF
True FLAG / FLAG_CATEGORY / FLAG_ROW elements	0 / 0 / 0
DATA nonfinite samples / negative-zero components	0 / 0
Source physical file lengths	409,594,838 bytes (65 files)

The logical DATA size is

$$N_v = N_r N_f N_p = 62,128 \times 192 \times 4 = 47,714,304, \qquad S_{\mathrm{DATA}} = 8N_v = 381,714,432 \text{ bytes}.$$

Logical arrays	Bytes
DATA	381,714,432
FLAG + FLAG_CATEGORY	95,428,608
WEIGHT + SIGMA	1,988,096
UVW	1,491,072
Other MAIN columns, including FLAG_ROW	4,783,856
Subtables	91,133
Total logical arrays	485,497,197

Logical bytes are not per-column disk usage: unpacked Boolean arrays count one byte per value here. Casacore separates logical tables from storage managers and stores Boolean values as bits. This MS uses shared StandardStMan files, so exact physical bytes cannot be attributed to individual columns by file lengths alone. Casacore Table Data System

UVW occupies one vector per row, not one per channel. Of 62,128 original vectors, 61,777 are bitwise distinct. A plain float64 dictionary plus uint32 row map would cost 1,731,160 logical bytes, exceeding the original 1,491,072 bytes; it is not selected on that basis. There is no tolerance-based geometry merging. For imaging one can form wavelength coordinates

$$\boldsymbol{u}_{\lambda}(r,f) = \boldsymbol{u}_{\mathrm{m}}(r)\,\frac{\nu_f}{c},$$

but this frequency-dependent conversion is not used for archival reconstruction.

3. Grouping: deciding which samples share a scale

3.1 Valid statistics and row identity

Let V_{rfp} be the complex64 visibility at original row r, channel f and correlation p. The supported quantized layout has a single integration and configuration, uniform cells and no duplicate row keys. Row identity includes the ordered antenna pair and distinguishing time/feed/field/ data-description/ scan/array/observation/processor/state context; antenna pairs are never swapped and visibilities are never conjugated to canonicalize them. Unsupported quantized layouts fail preflight. The exact archive path supports additional irregular cell layouts; those are not silently normalized into the current quantized path.

Define the usable-channel set and grouping statistic as

$$\mathcal{F}_{rp} = \big\{ f : \neg\mathrm{FLAG}_{rfp}, \neg\mathrm{FLAG_ROW}_{r}, \Re V_{rfp}, \Im V_{rfp} \ \text{ finite} \big\}, \qquad n_{rp} = |\ \mathcal{F}_{rp}\ |,$$

$$A_{rp} = \mathrm{median}_{f\in\mathcal{F}_{rp}}\ \mathrm{hypot}\ (\Re V_{rfp}, \Im V_{rfp}).$$

The magnitude calculation is performed robustly in float64. No-valid-channel rows receive a deterministic special class and stored valid count; NaNs cannot randomly order them. An absent record is not synthesized as an all-flagged row.

3.2 Permutation and block boundaries

For amplitude ordering, sort each product independently by auto/cross class, valid/no-valid class, A_{rp} and original row index. The current default `shared_order: false` allows different products to have different permutations. Let $\pi_p(j)$ be the original row at archived position j:

$$V^{\mathrm{arch}}_{jfp} = V_{\pi_p(j),f,p}, \qquad \hat{V}_{\pi_p(j),f,p} = \mathrm{Decode}\ (\mathrm{archive})_{jfp}.$$

Only baseline IDs/spectra are reordered. Channel order within a spectrum is unchanged. Within each class, consecutive groups of at most B rows and frequency segments of at most F channels define blocks

$$\mathcal{B}_{gsp} = \big\{ (\pi_p(j), f, p) : j_g \le j < j_{g+1}, f_s \le f < f_{s+1} \big\}.$$

Store actual permutations and class/group/segment boundaries; reconstruction does not rerun the sorting algorithm. Groups do not cross auto/cross, no-valid, bucket or configured geometry/antenna boundaries. Partial blocks retain exact bounds. A permutation alone provides no quantization gain if it leaves block membership unchanged.

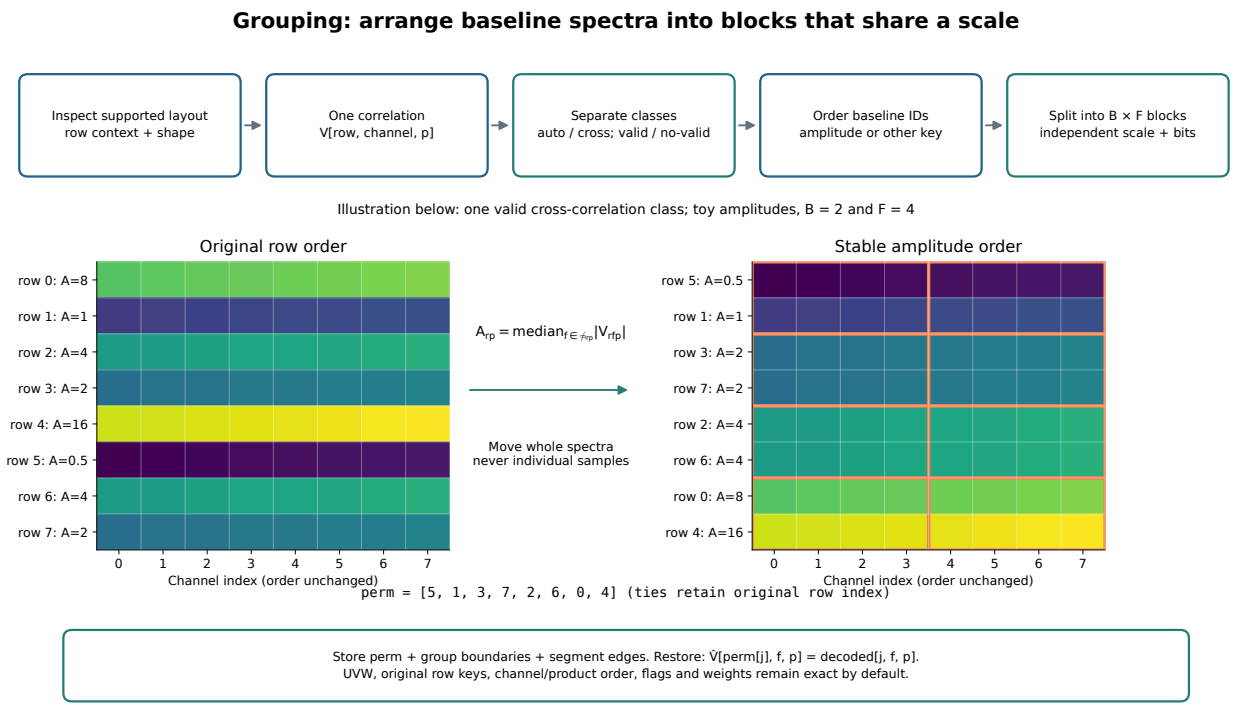


Figure 1. Illustrative grouping within one valid cross-correlation class, with toy amplitudes and $B = 2, F = 4$. Whole spectra move together. Orange rectangles are quantization blocks, not averaging bins. SVG, PDF.

For the measured $B = 128, F = 192$ configuration, all source rows have valid samples:

$$N_{\text{blocks}} = 4\left(\left\lceil \frac{352}{128} \right\rceil + \left\lceil \frac{61,776}{128} \right\rceil\right) = 4(3 + 483) = 1,944.$$

The final auto and cross groups contain 96 and 80 rows respectively. The four uint32 permutations occupy 994,048 logical bytes, or **488,354 stored bytes** in the measured archive. Their cost is included in all whole-archive ratios.

3.3 Available controls and actual defaults

Setting	Ordering	Maximum block, rows × channels × products
Quantizer defaults, when enabled	Original row order	1 × 192 × 1
Engineering example	Original row order	32 × 64 × 1
New-ladder measured profile	Independent amplitude ordering	128 × 192 × 1

Default **encoding mode remains lossless**. Other grouping controls include class-preserving original order, baseline length $\ell_r = \| \text{UVW}_r \|_2$, projected UV orientation $\theta_r = \text{atan2}\,(v_r, u_r) \bmod 2\pi$, combined length/orientation, and either ordered antenna endpoint. Orientation uses 24 bins by default (15° each), without folding opposite ordered baselines. Log-amplitude buckets use

$$k_{rp} = \left\lfloor \log_2 A_{rp} - \log_2 A_{\text{ref}} \right\rfloor, \quad A_{rp} > 0,$$

with zero and no-valid classes kept separate. The reference and actual bucket boundaries are stored. Combined geometry uses analogous factor-of-two radial bins referenced to 1 m. Amplitude statistics currently span the whole spectrum, even when scales use shorter frequency segments. There is no separately sorted per-frequency-segment mapping in the measured implementation.

4. Local uniform quantization and exact exceptions

4.1 Scale candidates

For one block with N complex samples, exclude flags, nonfinite values and signed-zero-protected samples from scale estimation. For the remaining samples, use the maximum absolute component

$$a_i = \max(|\,\Re V_i\,|, |\,\Im V_i\,|), \qquad t_q = Q_q(\{a_i\}), \qquad q \in \{1, 0.995\}.$$

This is a component-range statistic, distinct from the median complex magnitude used to order baselines. For a candidate bit width $b \in \{4, 8, 12, 16\}$, define

$$M_b = 2^{b-1} - 1, \qquad \Delta_{bq} = \mathrm{ceil}_{\mathrm{float32}}\big(t_q / M_b\big).$$

Here the ceiling means rounding to float32 and advancing to the next positive float32 if necessary so $M_b \Delta_{bq} \geq t_q$; it is not integer ceiling. The quantizer's stored scale is the **step** Δ, shared by real/imaginary components within that block. It is neither quantized again nor inferred at decode time. Zero-range blocks use $\Delta = 0$ explicitly. No complex offset or predictor is used.

4.2 Codes and exact decoding

For a finite, unprotected component x inside the supported interval, nearest ties-to-even rounding in float64 gives

$$q_b(x) = \mathrm{rint}_{\mathrm{even}}(x/\Delta) + M_b, \qquad q_b \in \{0, \ldots, 2M_b\}.$$

Code $2^b - 1$ is reserved and rejected. Ordinary zero is represented exactly by code M_b; zero-step blocks assign this code to ordinary zeros. The decoder is defined operationally, including storage precision:

$$\hat{x} = \mathrm{float32}\ [\mathrm{float32}\ (\mathrm{int32}\ (q_b) - M_b) \times \Delta_{\mathrm{stored}}].$$

Real and imaginary float32 bit patterns are interleaved into complex64. Evaluating the actual decoder avoids assuming ideal-arithmetic error bounds survive float32 scale storage and reconstruction. A 16-bit code is a scaled integer, **not IEEE float16** and not lossless float32 storage. Twelve-bit codes actually pack into 12 bits; a complex sample then has a three-byte code payload.

Within a block, samples use row-major baseline/channel order and each sample's real code precedes its imaginary code. Code bits are written least-significant bit first into a continuous byte stream, with zero padding only at the end. The header records the expected complex-sample count. Thus 4/8/12/16 component bits require 1/2/3/4 code bytes per complex sample before exceptions and headers.

4.3 Exceptions, flags and exact fallback

Every flagged sample, nonfinite sample, signed zero that would otherwise change, and sample outside $[-M_b\Delta, +M_b\Delta]$ in either component is restored from its original complex64 bits. The exception record stores a uint32 sample position and eight original bytes. Placeholders are quantized safely, then exact exceptions replace complete complex samples. Unflagged values are never silently saturated. Conventional lossless compression competes with an uncompressed body.

The default preserves MAIN flags, weights and auxiliary values exactly. An explicitly destructive `burn_flag` option omits original MAIN flag values and exports defined flag cells as false; it does not rerun a flagger. Subtable flags and visibility values remain protected. It was **disabled in all precision and imaging comparisons in this paper**. Since the current source flags are constant, burning them saved only 13,008 whole-archive bytes in a separate lossless test, not their approximately 95 MB of unpacked logical size. Flag policy and measurements

4.4 Full-range unsigned alternative

The zero-preserving codec uses $2^b - 1$ of the 2^b unsigned codes: the unused capacity is **one code**, not a separate sign bit. Setting $M_b = 2^b - 1$ and then using codes $0, ..., 2M_b - 1$ would require $b + 1$ total component bits (30 codes at nominal $b = 4$). A corrected full-range mapping, implemented separately as `uniform-midrise-v1`, defines

$$L_b = 2^b, \qquad C_b = \frac{L_b - 1}{2}, \qquad \Delta_{bq} = \mathrm{ceil}_{\mathrm{float32}}(t_q / C_b),$$

$$q_b(x) = \mathrm{rint}_{\mathrm{even}}(x/\Delta_{bq} + C_b), \qquad q_b \in \{0, ..., L_b - 1\},$$

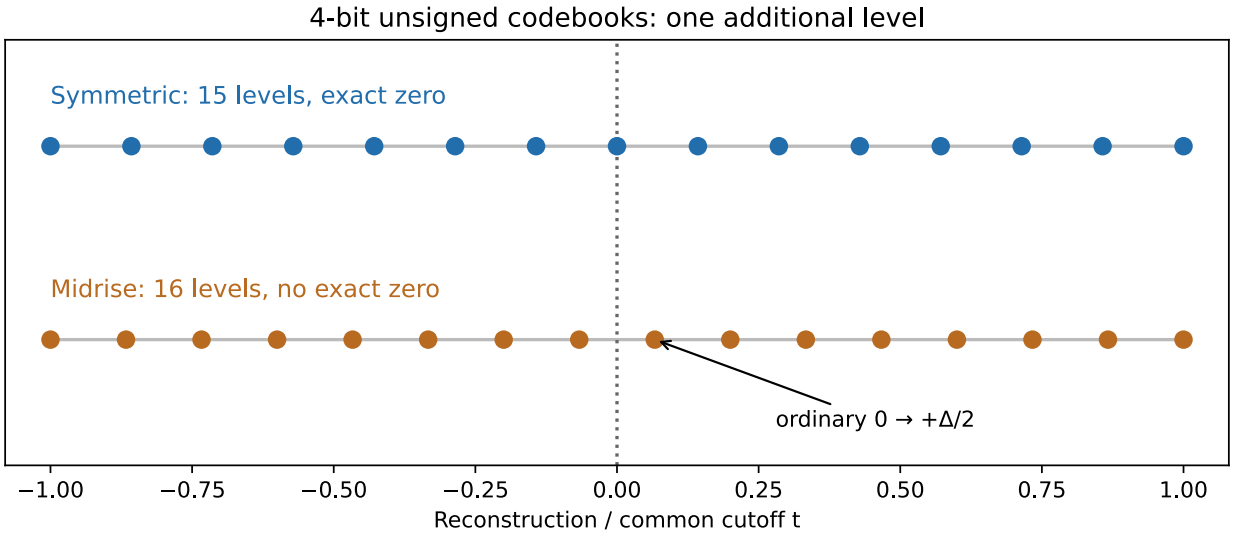
$$\hat{x} = \mathrm{float32}\,[\mathrm{float32}\,(\mathrm{float32}\,(q_b) - \mathrm{float32}\,(C_b))\Delta_{\mathrm{stored}}].$$

The exact half-integer C_b is determined by the bit width; it costs no stored offset. Outliers outside $[-C_b\Delta, +C_b\Delta]$ remain exact exceptions. Positive ordinary zero in a nonzero-step block reconstructs as $+\Delta/2$ under nearest-even rounding. Negative-zero samples and other protected values still retain their original bits; all-zero blocks use a zero step and remain exact. Ordinary-zero loss is thus explicit and opt-in. Stochastic rounding can remove its bias in expectation, but does not eliminate the added zero-component variance; the measurements below use nearest-even.

At the same cutoff, ignoring float32 rounding,

$$\frac{\Delta_{\mathrm{midrise}}}{\Delta_{\mathrm{symmetric}}} = \frac{2^b - 2}{2^b - 1}.$$

The high-resolution uniform-error approximation predicts a squared-error ratio equal to the square of this step ratio, approximately 0.8711 at 4 bits and 0.9922 at 8 bits. This approximation is not an error guarantee for structured visibilities, especially zero-rich autocorrelations. Both codecs evaluate the actual reconstruction against the same configured bounds and exact fallback.



Schematic: both codebooks use four total bits per component. The full-range alternative adds one level while losing the zero level. SVG, PDF.

5. Choosing precision by measured error and actual bytes

Let $\mathcal{U}$ be the block's source-unflagged finite samples and $e_i = \hat{V}_i - V_i$. Error calculations use float64 component differences:

$$\mathrm{MSE}_{\mathrm{comp}} = \frac{1}{2\mid \mathcal{U} \mid}\sum_{i\in\mathcal{U}}\mid e_i \mid^2, \quad \mu_e = \frac{1}{\mid \mathcal{U} \mid}\sum_{i\in\mathcal{U}} e_i, \quad E_{\max} = \max_{i\in\mathcal{U}}\mid e_i \mid .$$

Per-baseline MSE uses the same expression restricted to each row's valid channels. Protected values are checked for bitwise equality independently. Empty valid domains have zero measured numeric error; their protected values must still be exact. Real/imaginary errors and percentiles are also reported.

Check in the measured configuration	**Ceiling**
Block/whole-product component MSE	10^{-7}
Maximum within-block per-baseline component MSE	2×10^{-7}
Maximum complex absolute error	0.003
Absolute complex mean bias	0.0003
Optional p99 constraint	Not enabled; p99 still reported

For a quantized candidate with E exact exceptions, the uncompressed body size is

$$L_{bq} = \left\lceil \frac{2Nb}{8} \right\rceil + 12E.$$

With a 16-byte block header and the measured configuration's zlib-wrapped DEFLATE compressor (including its wrapper in the byte count),

$$C_{bq} = 16 + \min\big(\ L_{bq}, \mathrm{bytes}\ (\mathrm{DEFLATE}\ (\mathrm{body}_{bq}))\ \big)\ .$$

The exact candidate has body size $8N$ and undergoes the same compressed/raw comparison. Position indexes and CRCs have fixed size across candidates for a given block layout. Including them does not change which candidate wins, but they remain counted in archive totals. Let $\mathcal{A}$ contain candidates whose finite reconstruction and all configured error checks pass, including exact storage. Selection is

$$c^* = \arg\min_{c\in\mathcal{A}} C(c), \qquad c = (b, q)\ \ \text{or exact float32}\ .$$

Exact float32 is initialized as the incumbent; only a strictly smaller valid candidate replaces it. This is a local serialized-size choice, not a claim of global optimality after HDF5 overhead or across alternative groupings. Candidates already larger than the incumbent can skip decoder/error evaluation. The encoder then checks whole-product bounds before publishing a complete archive.

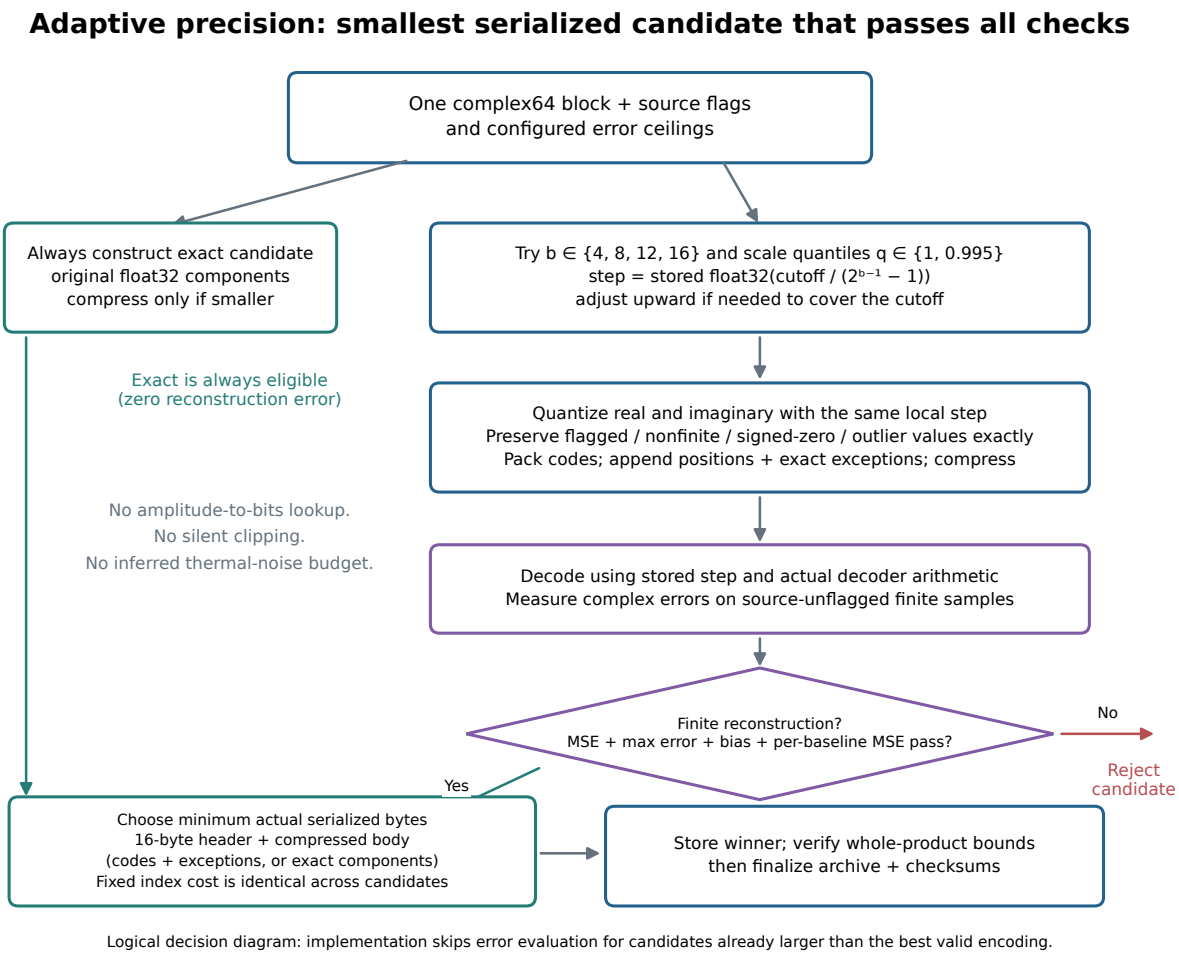


Figure 2. Logical decision path. Every block has its own scale and precision. The exact branch remains available even at 16 bits. SVG, PDF.

There is **no validated thermal-noise model** for this test. Were a defensible per-component thermal variance σ_{th}^2 supplied, one could require $\mathrm{MSE}_{\mathrm{comp}} \leq \epsilon\sigma_{\mathrm{th}}^2$. Independent, unbiased added noise would then increase RMS by $\sqrt{1+\epsilon}$. Those assumptions are not established here; median signal amplitude, image RMS and stored WEIGHT/SIGMA are not silently substituted for thermal variance. Original weights are unchanged.

6. Archive reconstruction and verification contract

The scientific schema is stored in HDF5, separate from the block codec. The manifest holds original descriptors/keywords/references, configuration, typed metadata, reconstruction maps, software hashes and inventory. Numeric payloads are portable little-endian values, not pickle objects. The decoder needs no network models or downloaded codebooks. Original supported coordinates and auxiliary columns are exact; selected quantized visibility columns are not.

I/O chunks and quantization blocks differ: independently compressed blocks are indexed in approximately 1 MiB visibility streams; ordinary HDF5 array chunks target 4 MiB. CRC32 per visibility block, HDF5 chunk checksums, dataset SHA-256 and manifest SHA-256 detect corruption. These are not authentication signatures. Source files are never overwritten. Incomplete archives/MSs remain distinguishable from published complete outputs.

MS export reconstructs original table relationships and cell/row/product/channel order, but does not reproduce original physical casacore storage files. Source and export comparison checked coordinates, metadata, flags and weights exactly; decoded DATA matched encode-time reconstructed errors. The output reopened with python-casacore and was imaged with

WSClean. These are specific checks, not a claim that every CASA imaging/calibration workflow is compatible. Format contract, verification evidence

7. Compression and grouping results

The earlier 14-profile sweep used **2/4/6/8** bits per component. The current default ladder is **4/8/12/16**; only the previous 128×192 amplitude configuration has been remeasured across ladders, not all 14 configurations. Historical report configurations are authoritative even though example configuration files have since changed. All comparisons below preserve flags and auxiliary columns.

Profile	Ladder	Whole MB	Mean component MSE
Exact archive, structural deduplication	Exact	331.220	0
Per baseline, 64-channel segments	2/4/6/8	155.010	3.211×10^{-8}
Class-original groups 32×64	2/4/6/8	173.188	3.530×10^{-8}
Baseline-length groups 32×64	2/4/6/8	163.285	3.902×10^{-8}
Amplitude buckets 32×64	2/4/6/8	157.822	3.675×10^{-8}
Amplitude sorting 32×64	2/4/6/8	146.699	3.940×10^{-8}
Amplitude sorting 128×192	2/4/6/8	136.921	2.908×10^{-8}
Amplitude sorting 128×192	4/8/12/16	**106.730**	2.812×10^{-8}

Common ceilings are not equal measured error. For example, sorted 32×64 blocks occupy fewer whole-archive bytes than the per-baseline control but have higher MSE. In that older comparison, serialized visibility blocks were actually 137.654 MB versus 137.313 MB; reduced headers/indexes account for much of the whole-archive advantage. Therefore the results do not establish that grouping intrinsically beats per-baseline scaling at equal error.

The no-dedup exact control occupied 331,693,912 bytes: structural deduplication saved 473,494 bytes separately from visibility quantization. Comparing ladders at the same 128×192 grouping saved **30,190,836 bytes (22.05%)**: new 12/16-bit candidates replaced earlier exact blocks. Larger individual code widths are not intrinsically smaller; they reduce the need for eight-byte complex fallback.

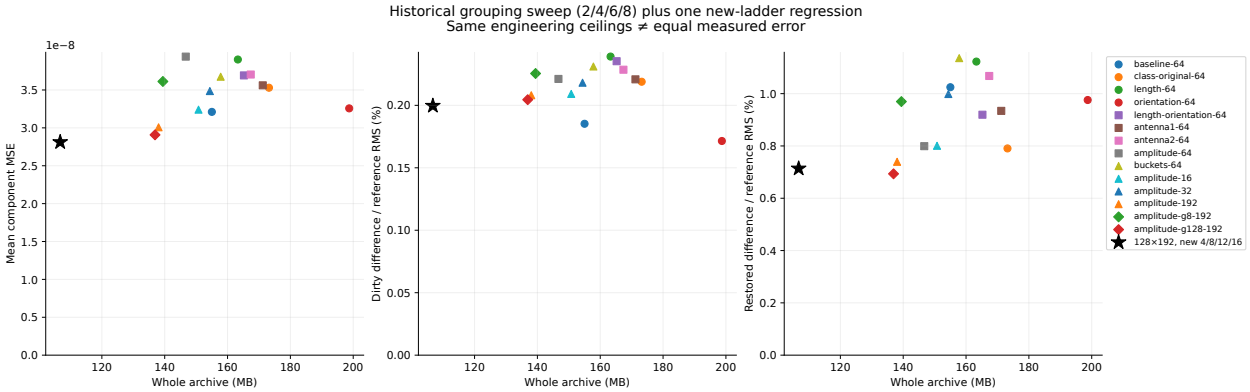


Figure 3. Fourteen historical grouping profiles and one new-ladder regression. All points include metadata, exceptions, maps and quality arrays. Points are not connected or interpolated into an equal-error curve. Lossless storage is 331.220 MB and lies outside the displayed lossy range. SVG.

Selected block precision	Blocks	Complex samples	Sample share
4	38	933,888	1.957%
8	1,556	38,240,256	80.144%
12	344	8,404,992	17.615%
16	6	135,168	0.283%
Exact float32 components	0	0	0%

The exact escape remains implemented and tested even though no entire block required it in this run. Quantized blocks can still contain exact exceptions; the table assigns samples by their block's precision, not the width of exception values. The whole archive is 106,730,115 bytes, with this measured partition:

Stored category	Bytes
Visibility block payload, including exceptions/headers	99,615,100
Visibility quality arrays	3,350,424
Grouping amplitude statistics and valid counts	1,676,751
Coordinates	988,098
Permutations	488,354
Visibility block indexes	25,572
Flags/weights; auxiliary columns; subtables	98; 184; 9,944
Container and manifest overhead	575,590

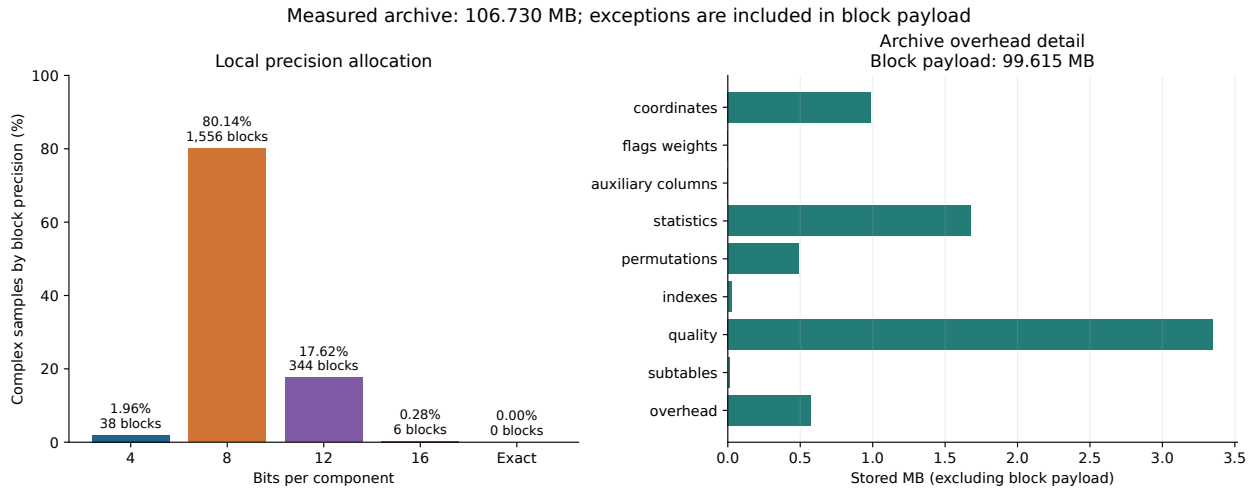


Figure 4. Local precision allocation and non-payload byte categories. Left bars show sample shares; labels also give block counts, which differ because of edge blocks. Right panel excludes the 99.615 MB visibility payload to show smaller categories. SVG.

Visibility-associated datasets including maps/statistics occupy 105,156,201 bytes, giving **3.630×** against logical complex64 DATA. The **3.838×** whole-MS ratio uses the physical source size and whole archive; these denominators are different and must not be exchanged.

8. Image-domain validation

WSClean 3.7 (`v3.7-79-g2d5c1ed8`) ran in the recorded local container `sha256:03ffbf70c829e220707b9843e96765f5886005b4b05a26344a0b5bb4d9ece8c6`. Both reference and decoded DATA used 2048×2048 pixels, 4 arcmin/pixel, Stokes I, Briggs robustness 0, multiscale CLEAN, mgain 0.8, auto-mask 5, auto-threshold 1, one thread and a 10,000-iteration limit. A common restoring beam was imposed: 666.0489742 × 517.0156755 arcsec, position angle 43.3244852°.

Both images hit the iteration cap, so CLEAN convergence is not claimed. The reference was repeated: single-thread/common-beam outputs were pixel-identical. Earlier eight-thread repeats showed about 0.682% restored-image RMS variation, motivating the single-thread controls. Automatic masks and nonlinear CLEAN paths can still differ when visibilities change. Imaging uses cross-correlations while the archive preserves all rows, including autos.

Let I be the reference image, $\hat{I}$ the decoded-data image and Ω the common finite, celestial-coordinate-valid pixel set. The metric is

$$R_I = \frac{\sqrt{|\ \Omega\ |^{-1} \sum_{x\in\Omega} (\hat{I}_x - I_x)^2}}{\sqrt{|\ \Omega\ |^{-1} \sum_{x\in\Omega} I_x^2}}.$$

This denominator contains sky signal and imaging artifacts; R_I is **not** an added-thermal-noise fraction. There are 2,320,513 valid pixels; 1,873,791 SIN projection-corner pixels are excluded. The mask is not a primary-beam or solar region selection. FITS units are labeled Jy/beam by software; absolute flux calibration has not been independently established.

Product	**Earlier ladder** $100R_I$	**New ladder** $100R_I$
Dirty	0.2045%	0.1996%
Restored	0.6934%	0.7137%
Model	—	4.8563%
Residual	—	1.5335%
PSF	0%	0%

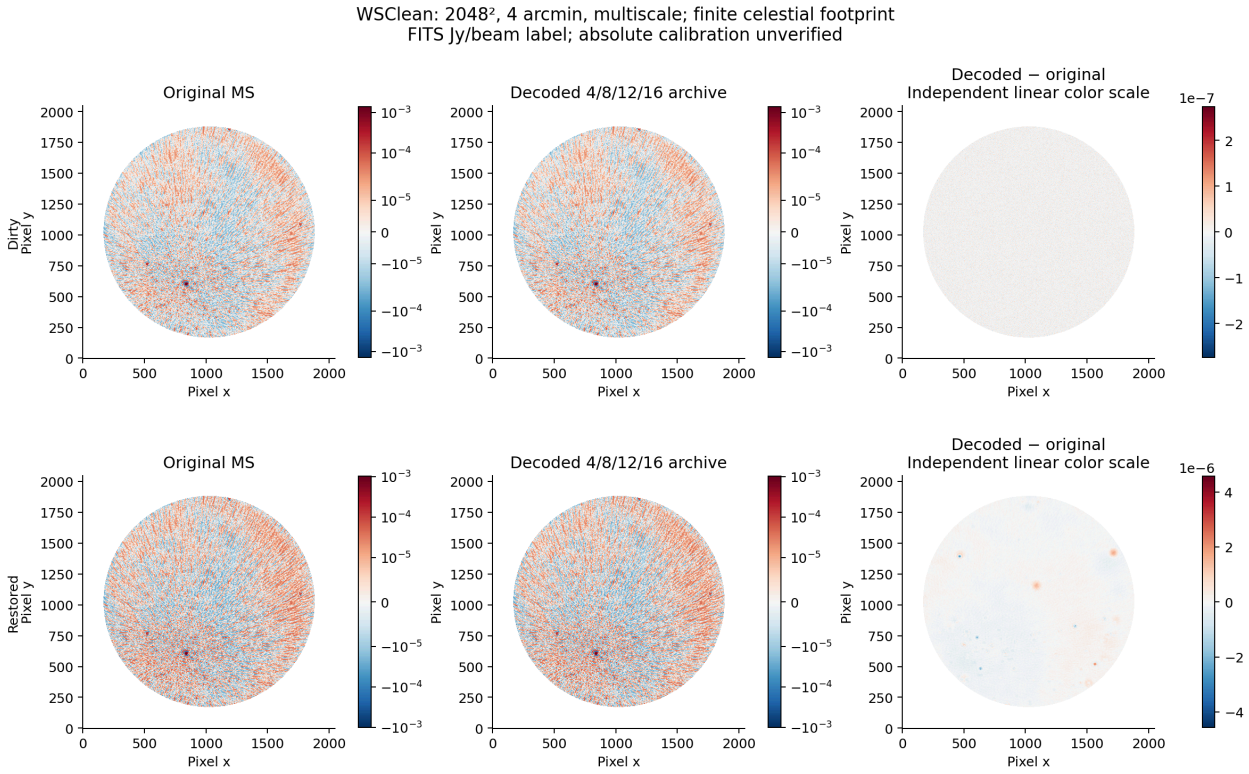


Figure 5. Real WSClean FITS data: dirty images (top) and restored images (bottom). Within each row the original/decoded panels share a symmetric-log scale; the difference panel uses its own linear scale. Color scales are explicit. No invalid projection-corner pixels are displayed.

The restored peak at fixed reference pixel `(y=599, x=839)` changes by −0.03976%, with zero peak-pixel displacement. The sum within a fixed four-pixel-radius reference aperture changes by −0.01177%. The brightest pixel has not been independently identified as the Sun; these are not calibrated solar flux results.

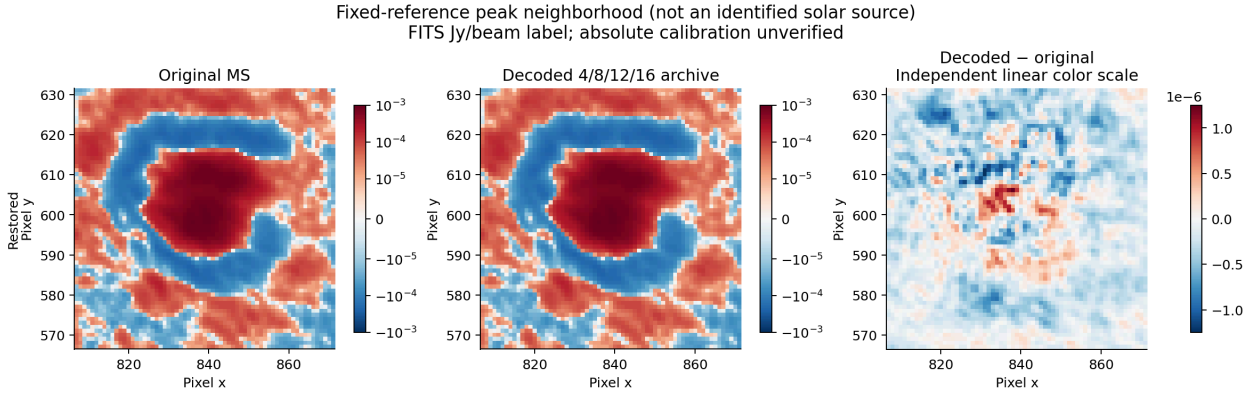


Figure 6. A 65×65-pixel neighborhood around the fixed restored-image reference peak. This zoom complements full-field RMS and is not a separate source-identification or weak-polarization validation.

The smaller new-ladder archive has slightly larger restored-image differences than the earlier ladder despite slightly smaller aggregate visibility MSE. This is a useful rate/distortion observation, not a universal monotonic relationship between visibility MSE and a nonlinear CLEAN product. Full per-region/peak/beam and checksum records are in the imaging report.

9. Encoding-speed benchmark

9.1 Test setup and timing scope

We measured the complete conversion of the inspected 409.595 MB MS into an archive using the amplitude-ordered 128-baseline × 192-channel configuration. The precision ladder was 4/8/12/16 bits per component, with the error ceilings in Section 5. Flags, weights, original coordinates and auxiliary columns were preserved.

The benchmark ran on an Intel Xeon Silver 4214 at 2.20 GHz with 48 logical CPUs and approximately 376.5 GiB RAM. The encoder used 24 CPU workers; OMP, OpenBLAS and MKL thread limits were set to one. The recorded software environment was Python 3.11.5, NumPy 1.26.4, python-casacore 3.5.2 and h5py 3.10.0.

Three runs were executed sequentially, each in a fresh process. The wall timer included program startup, full input inspection, compression and reconstruction-error checks, preservation of auxiliary data, archive writing, integrity checks and final publication. Source verification ran separately after each encode. The initial NAS copy/extraction, MS export and imaging were excluded from encode time.

The input had already been read, so the cache condition was warm/unspecified. The shared host had no CPU pinning, cache eviction or resource isolation. This task launched no other benchmark, profiling, test or imaging work during these three runs. Measurements therefore describe this host and cache condition.

9.2 Measured speed

Repeat	Complete encode process (s)	Separate verification (s)	Archive bytes
1	12.663	9.258	106,731,154
2	12.190	8.844	106,731,153
3	11.709	9.334	106,731,150

Median encode time was **12.190 s**, with range **11.709–12.663 s**. All three runs met the 15 s target. Mean time was 12.187 s and sample standard deviation was 0.477 s. Median throughput was **33.602 MB/s** of physical input size, equivalent to **3.914 million complex samples/s**. Median peak process memory was **1.573 GiB**.

The earlier one-worker encoder had a median of 170.620 s across three runs (range 168.688–181.006 s), using the same observation, grouping, error bounds and preservation policies. The current 24-worker benchmark has **14.00× lower median latency**. The worker counts and software versions differ, so this comparison measures the complete configurations. The earlier runs also overlapped figure generation and small synthetic tests; their shared-host conditions were less controlled. The original timings remain in the baseline report.

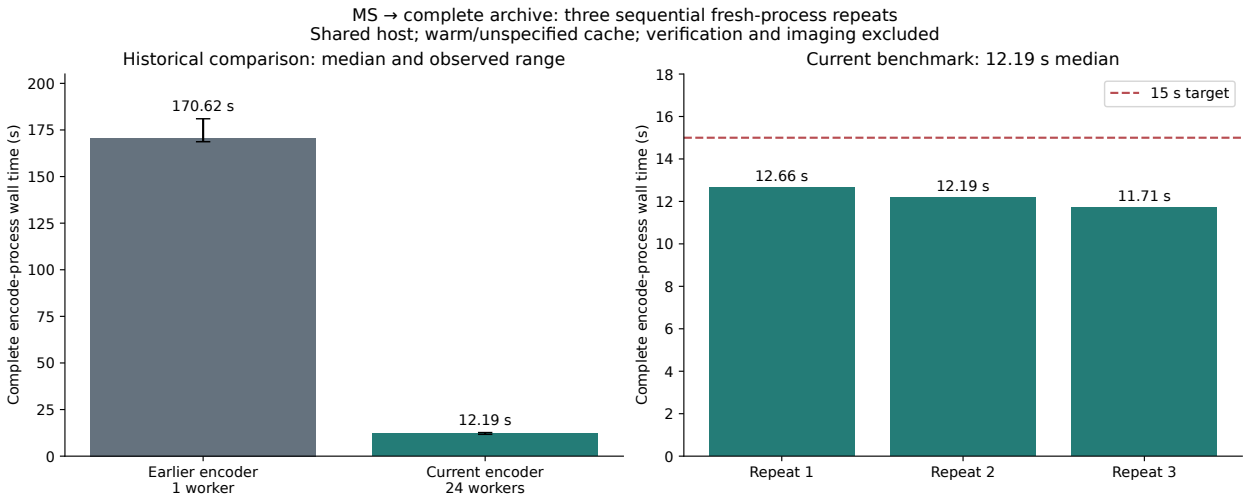


Figure 7. Complete MS-to-archive wall times. Left: median and observed minimum–maximum range for the earlier and current encoders, with three runs each. Right: individual current runs and the 15 s target. Verification and imaging are excluded from encode time. SVG, PDF.

9.3 Output validation and measurement limits

Every timed archive passed source verification. All inventoried datasets were identical across the three current runs; the four-byte spread in total file size comes from archive overhead. The median archive size was **106.731 MB**, retaining the approximately **3.838×** whole-MS reduction.

Compared with the earlier encoder, packed visibility payloads, maps, original coordinates, flags, weights and auxiliary values were bit-for-bit identical. Small differences in the numerical error summaries do not change the reconstructed visibilities. The scientific inputs to the existing WSClean comparison are unchanged; no additional imaging run is claimed. The second archive was exported to a new MS and passed independent source/export verification. All 65 source files still matched their frozen SHA-256 inventory.

These three runs establish an observed speed on this sample. Performance may differ with hardware, cache state, observation size, flag density and error bounds. The full configurations, per-run measurements and validation evidence are retained in the speed benchmark report and validation summary.

10. Exploratory two-level MX scaling

We also tested a smaller **32-baseline × 24-channel × 1-correlation megablock** with two levels of scaling. Each megablock holds 768 complex samples. For each channel, real and imaginary components separately form 32-baseline microblocks: 48 microblocks in total. The tested ordering is independent amplitude order, with the same auto/cross separation and exact row reconstruction as above.

The OCP MX specification defines FP4 E2M1, FP6 E2M3/E3M2 and FP8 E4M3/E5M2, with one UE8M0 power-of-two scale per 32 real elements. Its specified formats do not include MXFP16. Our 16-bit candidate is explicitly **custom scaled IEEE binary16**, with an exact float32 escape. A float32 outer factor and adaptive format tags make the overall record a custom extension. Related macroblock scaling has been studied for neural networks; those accuracy results do not establish visibility fidelity. Chhugani et al.

For element value c_{gmi} and micro exponent e_{gm}, reconstruction is

$$\hat{x}_{gmi} = \mathrm{float32}\left[\mathrm{float64}\,(S_g)\,\mathrm{ldexp}\,(\mathrm{float64}\,(c_{gmi}), e_{gm})\right].$$

The outer factor is either one or the maximum absolute unprotected component in the megablock, stored as float32. Micro exponents cover the corresponding component range. Candidates use nearest-even rounding and the original exact exception policy. Tests compare one format per megablock with a conservative per-microblock allocation. Every assembled candidate must pass the measured megablock and per-baseline error criteria, and compete in actual serialized bytes with exact storage. This is not an exhaustive search of all mixed allocations.

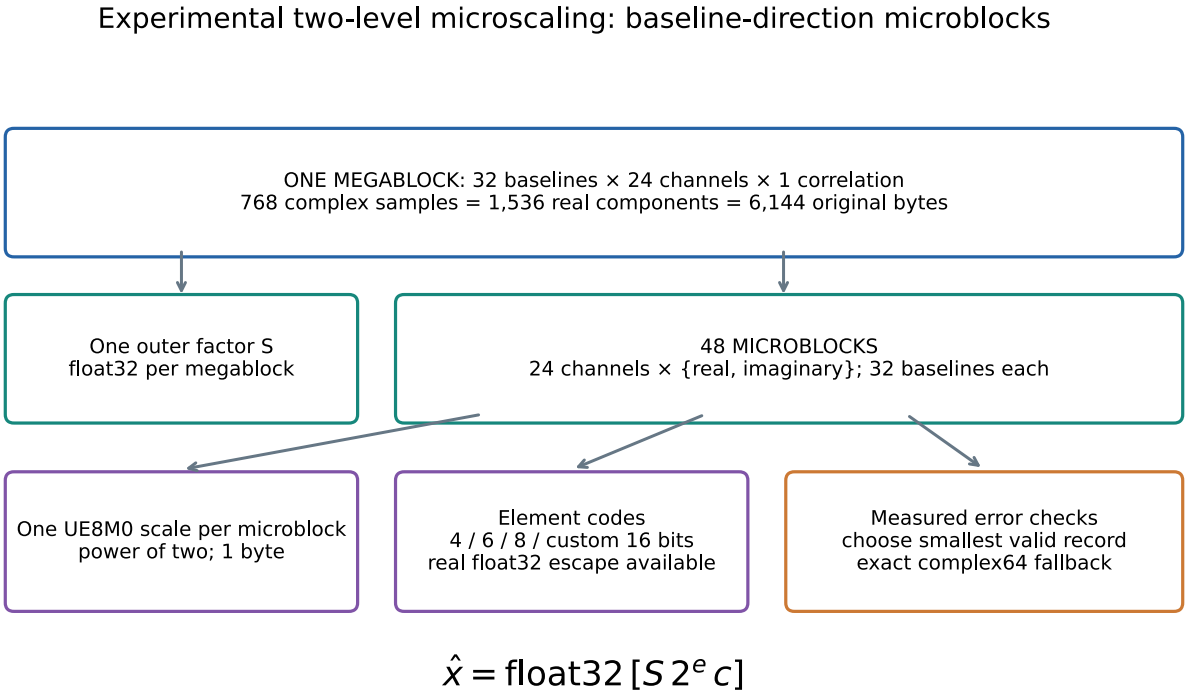


Figure 8. Experimental megablocks contain 48 independently scaled vectors of 32 real components. Real/imaginary components have separate micro scales. The outer factor is shared by the whole megablock. SVG.

All full runs cover the original 47,714,304 complex samples in 62,144 megablocks. The uniform controls use the same 32×24 grouping. The tighter control changes only the component/per-baseline MSE ceilings to $10^{-8}/(2 \times 10^{-8})$.

Codec / allocation	Serialized blocks MB	Component file MB	Component MSE
Uniform 4/8/12/16, original ceilings	101.569	107.935	3.1149×10^{-8}
MX with unit outer factor, mixed	172.260	179.268	5.0988×10^{-9}
MX with float32 outer factor, mixed	171.725	178.222	5.2093×10^{-9}
Uniform 4/8/12/16, tighter MSE ceilings	138.239	144.653	7.0993×10^{-10}

These component files are **not complete MS archives**. They hold DATA records, permutations, maps, quality arrays and configuration; MS export obtains exact auxiliary information from the original source. Serialized block bytes include headers, scales, tags and exceptions after actual secondary compression. The full-archive results in Sections 7 and 9 use a different 128×192 grouping and must not be compared as though these columns had the same preservation scope.

The tighter uniform control is **19.50% smaller** in serialized bytes than outer-scale MX, with **7.34× lower component MSE**. Its maximum complex error is 0.00025058 versus 0.00282696; maximum within-block per-baseline MSE and block bias are also lower. Adding the outer factor to MX saves only **0.311%** of serialized bytes relative to unit-factor MX, with slightly higher aggregate MSE.

Outer-scale MX selects 4/6/8/16/exact32 for **2.398/4.027/3.650/89.649/0.275%** of real components. Most components therefore need the custom FP16 rung under these absolute bounds. Code bytes dominate: 179.711 MB before secondary compression, compared with 2.975 MB of micro scales, 0.685 MB of format tags and 1.989 MB of headers. The outer factors are already included in those headers.

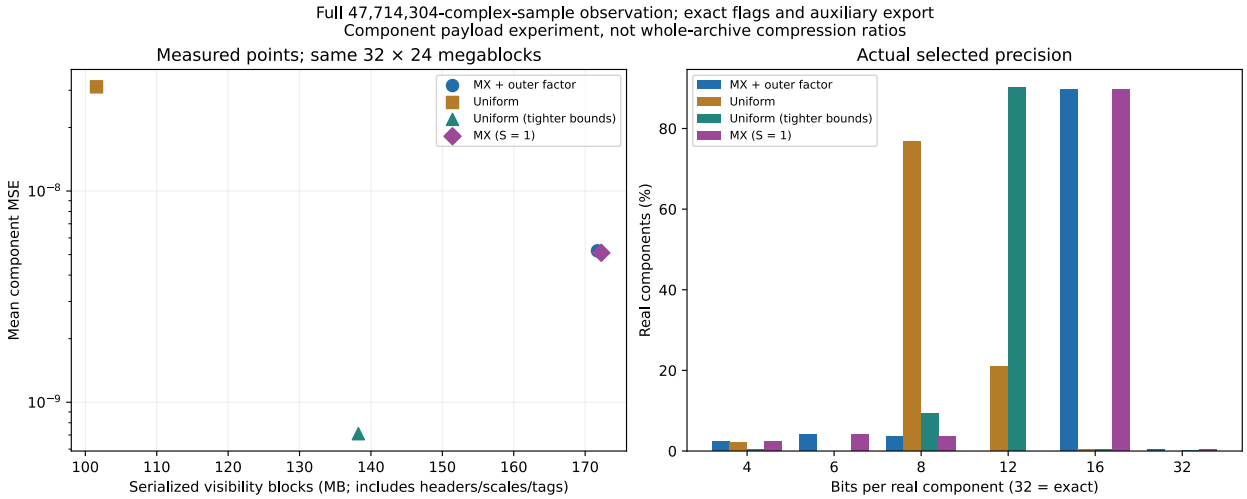


Figure 9. Measured full-data component rate–distortion points and precision shares at the same 32×24 block shape. Points are not interpolated into an equal-error curve. Exact32 denotes lossless components. SVG.

Single-worker diagnostic component-experiment times are 132.50 s for the uniform control, 165.05 s for unit-factor MX and 167.73 s for outer-factor MX. They include inspection, writing and per-block source-error checks; separate file decoding, MS export and imaging are excluded. Each is one warm/unspecified-cache run on the shared host. The tighter uniform run overlapped imaging and is not used for latency comparison. These scopes and worker counts differ from Section 9; the reference MX implementation has not demonstrated a complete archive within 15 s.

Matched WSClean 2048², 4-arcmin, multiscale Stokes-I imaging gives **0.06655% dirty** and **0.70078% restored** relative RMS differences over the valid celestial footprint. Model/residual differences are 4.93611%/1.56852%; the PSF is identical. Both images reach the 10,000-iteration cap, so convergence is not established. The tighter uniform control has not been imaged and no image-domain superiority is claimed for it. Thermal noise, calibration, weak polarization and temporal behavior remain unvalidated. Imaging evidence

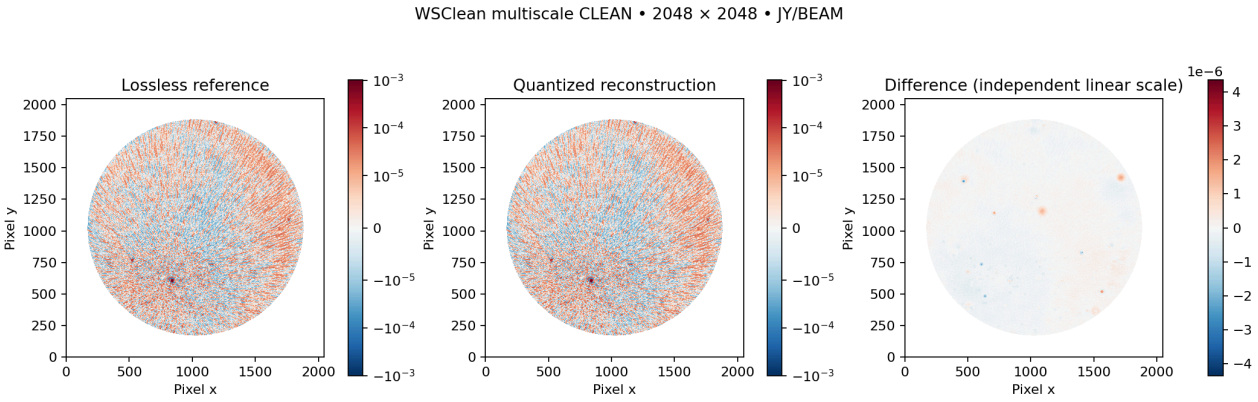


Figure 10. Original, outer-scale MX and difference restored images, using the same valid celestial footprint and matched restoring beam. Difference values have their own color scale; lower visibility MSE does not ensure proportionally smaller changes after nonlinear CLEAN.

The outer-scale MX export preserves all 132 checked non-DATA column entries and metadata exactly, including original UVW, flags and weights. A fresh component decode matches encode-time reconstruction bitwise. The experiment demonstrates that the proposed representation is feasible, but does not establish a storage advantage for these formats, scale rules and error ceilings. Details and evidence are in the microscaling note and machine-readable study.

11. Full-range unsigned-code comparison

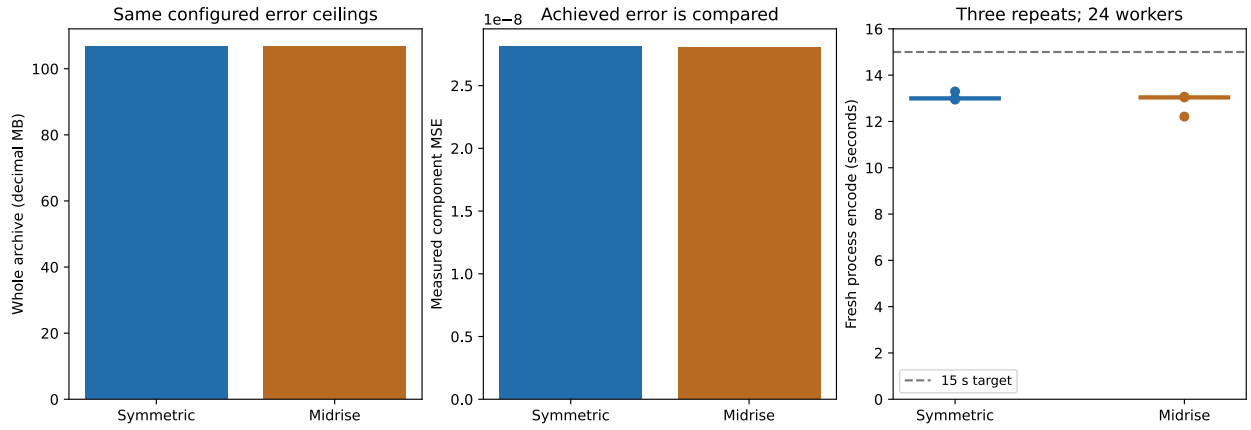
After the MX experiment, the scaled unsigned-integer route remains the primary implementation. Section 4.4's `uniform-midrise-v1` is an independently decodable alternative within the existing archive, with a distinct block mapping identifier. It does not add an exponent stream, complex offset or new container.

We compare it against `uniform-symmetric-v1` on the complete inspected MS with the same per-product amplitude ordering, 128×192×1 blocks, scale candidates, 4/8/12/16 ladder, error bounds and lossless auxiliary-column policies. The two codecs alternate in six sequential fresh-process runs, three each, with 24 workers and no concurrent imaging. Verification is separately timed and passes for all six outputs. The shared-host cache was not evicted; these are not cold-cache measurements. Each run includes startup, inspection, encoding, quality accounting, all archive writes and integrity publication, as in Section 9.

Measurement	Symmetric, zero-preserving	Full-range midrise
Whole archive, first repeat (MB)	106.731246	106.732004
Serialized visibility blocks (MB)	97.058796	97.081093
Aggregate component MSE	`2.811694574e-8`	`2.805425666e-8`
4-bit / 8-bit blocks	38 / 1,556	40 / 1,557
12-bit / 16-bit / exact blocks	344 / 6 / 0	341 / 4 / 2
Encode time, each repeat (s)	12.997 / 12.945 / 13.295	13.069 / 13.039 / 12.210
Median encode time (s)	12.997	13.039

There are 1,944 blocks per run. Scientific dataset hashes are identical across repetitions of each codec; small file-length variations reflect recorded timing and provenance metadata. The 22,297-byte increase in serialized visibility records fits in unchanged HDF5 payload allocation. The final archive is about 758 bytes larger in the first repeat; no storage advantage

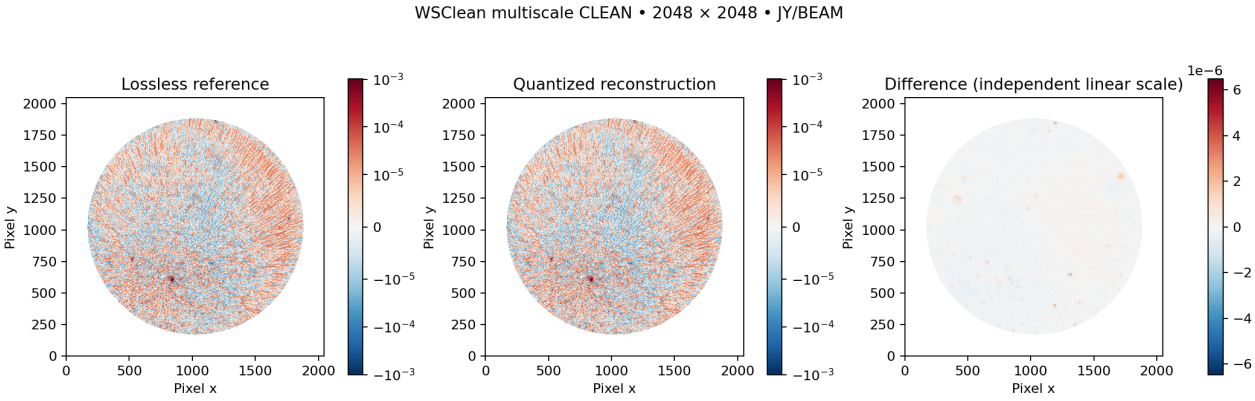
is demonstrated. Component MSE is 0.223% lower overall, but equal ceilings do not imply identical achieved error. Both codecs continue to meet the 15-second target in this experiment.



Measured full-MS comparison with identical configured error ceilings. Runtime points are separate processes; horizontal marks show medians. SVG, PDF.

The full-product XX/YY imaginary biases change from `-1.4032e-7` / `1.9476e-9` to `1.1083e-6` / `9.1734e-7`. This is consistent with the missing zero level in zero-rich parallel-hand autocorrelations. All configured bias/error checks still pass, with two exact-block escapes. The zero-preserving codebook therefore stays the default; the new mapping is an opt-in experiment, not a general improvement. Detailed byte categories, per-product and baseline-class errors, source hashes and reproducible configuration are in the unsigned-code report and development note.

The exported MS preserves all 132 checked non-DATA entries and metadata exactly, and all 65 source file hashes are unchanged. With the same pinned WSClean setup as Section 8 (2048^2, 4 arcmin, Briggs 0, multiscale, Stokes I, one worker, fixed reference beam), dirty/restored relative RMS differences are **0.1985% / 1.1142%**. Model/residual differences are 10.7263% / 2.4779%; the PSF difference is zero. The run reaches the 10,000-component cap, so CLEAN convergence is not established. The existing symmetric reconstruction's restored difference is 0.7137%; the small visibility-MSE improvement therefore does not imply improved CLEAN agreement. WSClean uses the 61,776 cross-correlation rows; this does not validate the auto-correlation imaginary offsets. Separate CASA, weak-polarization, calibration and temporal acceptance checks were not performed for this mapping.



Original and midrise restored Stokes I images and their actual difference, using the same beam. The difference panel has its own color scale. The matching image report records full masks and metrics.

12. Limits and conclusions

The results demonstrate a reconstructible prototype, not a final archive policy. Grouping can reduce whole-archive bytes, but overhead and unequal measured error matter; amplitude sorting has not been shown universally superior to buckets or per-baseline scaling. The new ladder reduced exact-block fallback for one tested grouping, not for an exhaustive new-ladder grid. Production scale metadata is not quantized; Section 10′s separate MX component experiment uses exponent micro scales.

Important unavailable checks remain: matched Dysco settings, calibrated thermal noise, weak Q/U/V products, calibration-solution changes, identified solar flux, narrow spectral features and multi-frame temporal behavior. One integration cannot establish transient/light-curve fidelity. No science-approved error ceilings have been selected. Stochastic rounding is implemented and tested, but these measurements use nearest-even rounding; subtractive dithering, NF4 codebooks and learned/model residual codecs are not implemented.

The practical conclusion is limited: independent local scaling, measured 4/8/12/16-bit selection, exact exceptions and exact-coordinate reconstruction provide a 106.730 MB archive for this 409.595 MB input under the stated engineering constraints. The original resolution and scientific context are retained; the quantized visibility precision is not. The default encoder remains lossless.

13. Reproduction and evidence

Install the validation dependencies and run from the repository root:

```
python -m pip install -e '.[validation]'
python -m unittest discover -s tests -v

# Full regression and MS export, using a NEW output directory:
visarchive benchmark INPUT.ms --matrix configs/benchmark-grouping.json \
  --experiments amplitude-g128-192 --output results/new-precision-run

# Speed benchmark: three fresh processes, 24 workers, verification separate.
OMP_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 MKL_NUM_THREADS=1 \
  visarchive benchmark INPUT.ms --matrix configs/benchmark-performance.json \
  --output results/new-performance-run
python scripts/summarize_runtime.py results/new-performance-run/results.json \
  --source-inventory reports/sample-copy-after-grouping.json \
  --output reports/new-runtime.json

# Diagrams/charts; image panels additionally require the local generated FITS files.
python scripts/build_paper_figures.py

# Matched unsigned-code comparison; use a fresh result directory.
OPENBLAS_NUM_THREADS=1 OMP_NUM_THREADS=1 visarchive benchmark INPUT.ms \
  --matrix configs/benchmark-midrise.json --output results/new-uint-comparison
python scripts/report_midrise.py --figures-only
```

Use `--force` only to regenerate existing paper figures. For a checkout without the untracked observation/FITS data, add `--skip-images` to regenerate diagrams and report-derived numerical charts. The source-inventory argument above belongs to this specific sample; use a matching inventory for another input. Historical reports retain old configurations and are not recreated simply by running today's default matrices. The measured preset's installation prerequisites are listed in the benchmark setup notes.

The WSClean reproduction command uses a private working MS copy:

```
python scripts/run_wsclean.py RESTORED.ms --output results/new-images \
  --container sha256:03ffbf70c829e220707b9843e96765f5886005b4b05a26344a0b5bb4d9ece8c6 \
  --size 2048 --scale 4amin --threads 1 \
  --beam-from results/wsclean-grouping/reference/image-image.fits
```

```
python scripts/compare_wsclean.py results/wsclean-grouping/reference-single \
  results/new-images --report reports/new-image-comparison.json
```

Primary local evidence:

- Input inventory
- Historical grouping measurements
- New-ladder configuration, byte/error accounting and imaging summary
- Full WSClean comparison
- Single-thread reference repeatability
- Historical timing measurements
- Current encoding-speed benchmark
- Output identity and independent verification
- Two-level MX study and component-byte accounting
- MX image-domain comparison
- Full-range unsigned-code comparison
- Full-range unsigned-code imaging
- Archive/codec format

All flow diagrams, measured charts and image panels are in figs/. Figures 1–2 are explicitly schematic. Numerical figures derive from the linked reports; image panels derive from the actual matched FITS outputs. Observation payloads, archives, exported MSs and FITS files remain outside Git.